RESULTS:

Terminal 1: Mininet:

Test Scenario 1 — Connectivity (Allowed vs Blocked)

```
*** Starting network
*** Configuring hosts
h1 h2 h3 h4 h5 h6
*** Starting controller
c0
*** Starting 3 switches
```

```
*** Waiting 3 s for controller to install flows …

*** Verifying OVS OpenFlow version

============================================================
  Topology ready. Try these commands in the CLI:

  pingall                      - full mesh ping
  h1 ping -c4 h6               - end-to-end ping
  h1 iperf -s &                - iperf server on h1
  h6 iperf -c 10.0.0.1 -t 10  - throughput test
  sh ovs-ofctl dump-flows s1   - show flow table

  # Simulate port-down event:
  link s1 s2 down              - bring inter-switch link down
  link s1 s2 up                - restore it
============================================================
*** Starting CLI:
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3 h4 h5 h6
h2 -> h1 h3 h4 h5 h6
h3 -> h1 h2 h4 h5 h6
h4 -> h1 h2 h3 h5 h6
h5 -> h1 h2 h3 h4 h6
h6 -> h1 h2 h3 h4 h5
*** Results: 0% dropped (30/30 received)
```

```
h6 -> h1 h2 h3 h4 h5
*** Results: 0% dropped (30/30 received)
mininet> h1 ping -c4 h6
PING 10.0.0.6 (10.0.0.6) 56(84) bytes of data.
64 bytes from 10.0.0.6: icmp_seq=1 ttl=64 time=12.3 ms
64 bytes from 10.0.0.6: icmp_seq=2 ttl=64 time=12.4 ms
64 bytes from 10.0.0.6: icmp_seq=3 ttl=64 time=12.3 ms
64 bytes from 10.0.0.6: icmp_seq=4 ttl=64 time=12.3 ms

--- 10.0.0.6 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3003ms
rtt min/avg/max/mdev = 12.304/12.320/12.356/0.020 ms
mininet> link s1 s2 down
mininet> h1 ping -c2 h4
PING 10.0.0.4 (10.0.0.4) 56(84) bytes of data.

--- 10.0.0.4 ping statistics ---
2 packets transmitted, 0 received, 100% packet loss, time 1025ms

mininet> link s1 s2 up
mininet> h1 ping -c4 h6
PING 10.0.0.6 (10.0.0.6) 56(84) bytes of data.
64 bytes from 10.0.0.6: icmp_seq=1 ttl=64 time=26.3 ms
64 bytes from 10.0.0.6: icmp_seq=2 ttl=64 time=14.5 ms
64 bytes from 10.0.0.6: icmp_seq=3 ttl=64 time=12.5 ms
64 bytes from 10.0.0.6: icmp_seq=4 ttl=64 time=12.4 ms

--- 10.0.0.6 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
```

```
64 bytes from 10.0.0.6: icmp_seq=4 ttl=64 time=12.4 ms

--- 10.0.0.6 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 12.397/16.402/26.281/5.763 ms
mininet> h1 iperf -s &
mininet> h6 iperf -c 10.0.0.1 -t 10
------------------------------------------------------------
Client connecting to 10.0.0.1, TCP port 5001
TCP window size: 85.3 KByte (default)
------------------------------------------------------------
[  1] local 10.0.0.6 port 51164 connected with 10.0.0.1 port 5001
[ ID] Interval       Transfer     Bandwidth
[  1] 0.00-11.34 sec   129 MBytes  95.1 Mbits/sec
mininet> quit

*** Stopping network
*** Stopping 1 controllers
c0
*** Stopping 8 links
........
*** Stopping 3 switches
s1 s2 s3
*** Stopping 6 hosts
h1 h2 h3 h4 h5 h6
*** Done
```

Test Scenario 2 — Port Failure (Normal vs Failure)

```
admin@fedora:~/Rohini/_SEM4/CN_LAB/Orange/sdn-port-monitor$ sudo ovs-ofctl -O OpenFlow13 dump-flows s1
sudo ovs-ofctl -O OpenFlow13 dump-ports s1
[sudo] password for admin:
 cookie=0x0, duration=365.002s, table=0, n_packets=285, n_bytes=59334, priority=0 actions=CONTROLLER:65535
OFPST_PORT reply (OF1.3) (xid=0x2): 5 ports
  port LOCAL: rx pkts=0, bytes=0, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=0, bytes=0, drop=0, errs=0, coll=0
           duration=365.126s
  port  "s1-eth4": rx pkts=273, bytes=58722, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=232, bytes=41258, drop=4, errs=0, coll=0
           duration=365.137s
  port  "s1-eth1": rx pkts=106, bytes=7660, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=327, bytes=83444, drop=0, errs=0, coll=0
           duration=365.138s
  port  "s1-eth2": rx pkts=87, bytes=6246, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=315, bytes=79156, drop=0, errs=0, coll=0
           duration=365.138s
  port  "s1-eth3": rx pkts=95, bytes=6674, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=321, bytes=79408, drop=0, errs=0, coll=0
           duration=365.134s
admin@fedora:~/Rohini/_SEM4/CN_LAB/Orange/sdn-port-monitor$
```

Terminal 2: OSKen Manager

OSKen instead of Ryu (since Ryu is no longer maintained and hence not supported in latest Python versions)

```
EOF
(sdn_env) admin@fedora:~/Rohini/_SEM4/CN_LAB/Orange/OSKen$ osken-manager /home/admin/Rohini/_SEM4/CN_LAB/Orange/sdn-port-monitor/a_port_monitor.py
loading app /home/admin/Rohini/_SEM4/CN_LAB/Orange/sdn-port-monitor/a_port_monitor.py
loading app os_ken.controller.ofp_handler
instantiating app /home/admin/Rohini/_SEM4/CN_LAB/Orange/sdn-port-monitor/a_port_monitor.py of PortMonitorController
PortMonitorController started. Log → /home/admin/Rohini/_SEM4/CN_LAB/Orange/sdn-port-monitor/../logs/port_events.log
instantiating app os_ken.controller.ofp_handler of OFPHandler
[SWITCH CONNECTED] dpid=0000000000000001
[SWITCH CONNECTED] dpid=0000000000000002
[SWITCH CONNECTED] dpid=0000000000000003
[PORT EVENT] {"timestamp": "2026-04-19 00:43:51.290410", "dpid": "0000000000000001", "port_no": 4, "port_name": "s1-eth4", "reason": "MODIFY", "state"
: "UP"}

============================================================
  *** ALERT: PORT STATUS CHANGE ***
  Time      : 2026-04-19 00:43:51
  Switch    : dpid=0000000000000001
  Port No   : 4
  Port Name : s1-eth4
  Reason    : MODIFY
  State     : UNKNOWN  →  UP
============================================================

[PORT EVENT] {"timestamp": "2026-04-19 00:43:51.292167", "dpid": "0000000000000001", "port_no": 4, "port_name": "s1-eth4", "reason": "MODIFY", "state"
: "DOWN"}

============================================================
  *** ALERT: PORT STATUS CHANGE ***
  Time      : 2026-04-19 00:43:51
  Switch    : dpid=0000000000000001
```

```
[PORT EVENT] {"timestamp": "2026-04-19 00:43:51.292167", "dpid": "0000000000000001", "port_no": 4, "port_name": "s1-eth4", "reason": "MODIFY", "state"
: "DOWN"}

============================================================
  *** ALERT: PORT STATUS CHANGE ***
  Time      : 2026-04-19 00:43:51
  Switch    : dpid=0000000000000001
  Port No   : 4
  Port Name : s1-eth4
  Reason    : MODIFY
  State     : UP  →  DOWN
============================================================

[PORT EVENT] {"timestamp": "2026-04-19 00:43:51.292972", "dpid": "0000000000000002", "port_no": 2, "port_name": "s2-eth2", "reason": "MODIFY", "state"
: "UP"}

============================================================
  *** ALERT: PORT STATUS CHANGE ***
  Time      : 2026-04-19 00:43:51
  Switch    : dpid=0000000000000002
  Port No   : 2
  Port Name : s2-eth2
  Reason    : MODIFY
  State     : UNKNOWN  →  UP
============================================================

[PORT EVENT] {"timestamp": "2026-04-19 00:43:51.293716", "dpid": "0000000000000002", "port_no": 2, "port_name": "s2-eth2", "reason": "MODIFY", "state"
: "DOWN"}
```

```
[PORT EVENT] {"timestamp": "2026-04-19 00:43:51.293716", "dpid": "0000000000000002", "port_no": 2, "port_name": "s2-eth2", "reason": "MODIFY", "state"
: "DOWN"}

============================================================
  *** ALERT: PORT STATUS CHANGE ***
  Time      : 2026-04-19 00:43:51
  Switch    : dpid=0000000000000002
  Port No   : 2
  Port Name : s2-eth2
  Reason    : MODIFY
  State     : UP  →  DOWN
============================================================

[PORT EVENT] {"timestamp": "2026-04-19 00:44:59.602319", "dpid": "0000000000000002", "port_no": 2, "port_name": "s2-eth2", "reason": "MODIFY", "state"
: "UP"}

============================================================
  *** ALERT: PORT STATUS CHANGE ***
  Time      : 2026-04-19 00:44:59
  Switch    : dpid=0000000000000002
  Port No   : 2
  Port Name : s2-eth2
  Reason    : MODIFY
  State     : DOWN  →  UP
============================================================

[PORT EVENT] {"timestamp": "2026-04-19 00:44:59.603074", "dpid": "0000000000000001", "port_no": 4, "port_name": "s1-eth4", "reason": "MODIFY", "state"
: "UP"}

============================================================
```

```
============================================================
  *** ALERT: PORT STATUS CHANGE ***
  Time      : 2026-04-19 00:44:59
  Switch    : dpid=0000000000000001
  Port No   : 4
  Port Name : s1-eth4
  Reason    : MODIFY
  State     : DOWN  →  UP
============================================================
```

```
Message: '[SWITCH DISCONNECTED] dpid=%016x'
Arguments: (None,)
[PORT EVENT] {"timestamp": "2026-04-19 00:52:55.072860", "dpid": "0000000000000001", "port_no": 1, "port_name": "s1-eth1", "reason": "MODIFY", "state"
: "UP"}
```

Terminal 3:
GUI

```
SDN Port Status Monitor Dashboard          2026-04-19 01:24:51

Switch dpid=0000000000000001
  Port  Name            State      RX Pkts     TX Pkts      RX Bytes      TX Bytes  Last Change
  ────  ──────────────  ─────  ──────────  ──────────  ────────────  ────────────  ───────────────────
     1  s1-eth1         UP              0           0             0             0  2026-04-19 01:04:43
     2  s1-eth2         UP              0           0             0             0  2026-04-19 01:04:43
     3  s1-eth3         UP              0           0             0             0  2026-04-19 01:04:43
     4  s1-eth4         UP              0           0             0             0  2026-04-19 01:11:07

Switch dpid=0000000000000002
  Port  Name            State      RX Pkts     TX Pkts      RX Bytes      TX Bytes  Last Change
  ────  ──────────────  ─────  ──────────  ──────────  ────────────  ────────────  ───────────────────
     1  s2-eth1         UP              0           0             0             0  2026-04-19 01:04:43
     2  s2-eth2         UP              0           0             0             0  2026-04-19 01:11:07
     3  s2-eth3         UP              0           0             0             0  2026-04-19 01:00:32

Switch dpid=0000000000000003
  Port  Name            State      RX Pkts     TX Pkts      RX Bytes      TX Bytes  Last Change
  ────  ──────────────  ─────  ──────────  ──────────  ────────────  ────────────  ───────────────────
     1  s3-eth1         UP              0           0             0             0  2026-04-19 01:04:43
     2  s3-eth2         UP              0           0             0             0  2026-04-19 01:04:43
     3  s3-eth3         UP              0           0             0             0  2026-04-19 01:00:32

Recent Events (last 15)
──────────────────────────────────────────────────────────

Refreshing every 2s — Ctrl+C to exit
```

Terminal 4:
iperf:

Test Scenario 3 — Throughput (iperf)

mininet> h1 iperf -s &

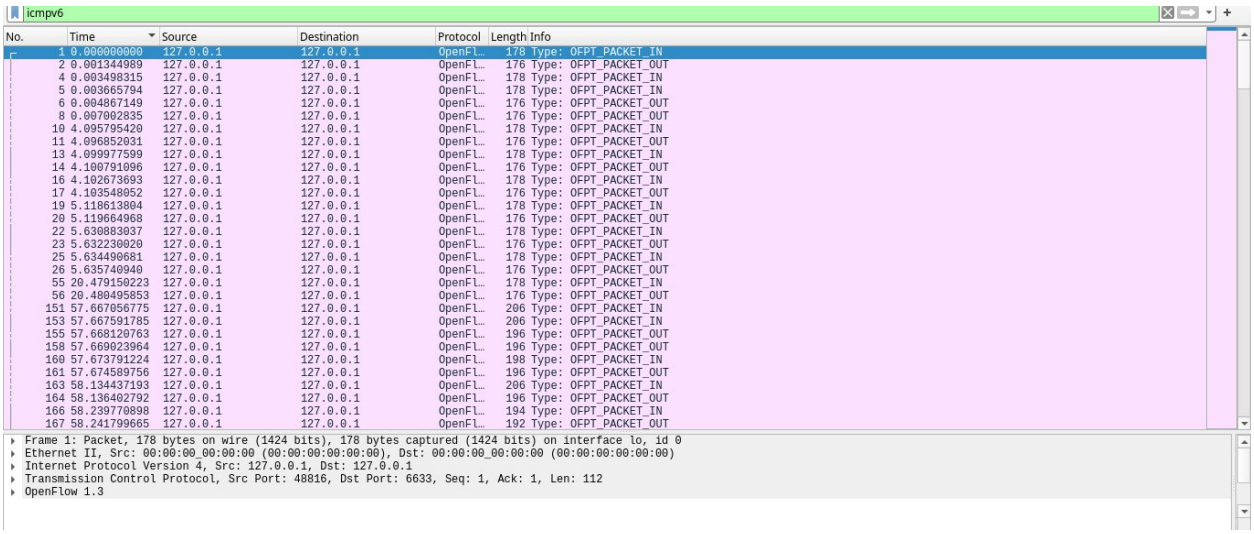
mininet> h6 iperf -c 10.0.0.1 -t 10

```
mininet> link s1 s2 up
mininet> h1 iperf -s &
[1] 23661
mininet> h6 iperf -c 10.0.0.1 -t 10
------------------------------------------------------------
Client connecting to 10.0.0.1, TCP port 5001
TCP window size: 85.3 KByte (default)
------------------------------------------------------------
[  1] local 10.0.0.6 port 46696 connected with 10.0.0.1 port 5001
[ ID] Interval       Transfer     Bandwidth
[  1] 0.00-11.01 sec   125 MBytes  95.2 Mbits/sec
mininet>
```

Wireshark:

– Scenario 1: Connectivity (Allowed vs. Blocked)

When it is down:

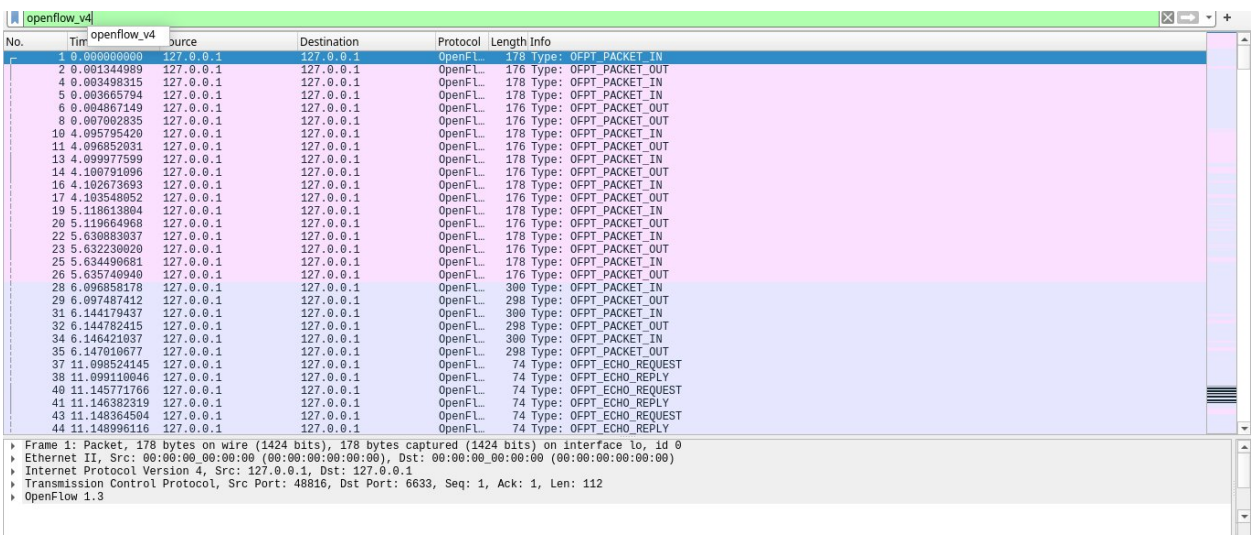


icmpv6

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	127.0.0.1	127.0.0.1	OpenFl…	178	Type: OFPT_PACKET_IN
2	0.001344989	127.0.0.1	127.0.0.1	OpenFl…	176	Type: OFPT_PACKET_OUT
4	0.003498315	127.0.0.1	127.0.0.1	OpenFl…	178	Type: OFPT_PACKET_IN
5	0.003665794	127.0.0.1	127.0.0.1	OpenFl…	178	Type: OFPT_PACKET_IN
6	0.004867149	127.0.0.1	127.0.0.1	OpenFl…	176	Type: OFPT_PACKET_OUT
8	0.007002835	127.0.0.1	127.0.0.1	OpenFl…	176	Type: OFPT_PACKET_OUT
10	4.095795420	127.0.0.1	127.0.0.1	OpenFl…	178	Type: OFPT_PACKET_IN
11	4.096852031	127.0.0.1	127.0.0.1	OpenFl…	176	Type: OFPT_PACKET_OUT
13	4.099977599	127.0.0.1	127.0.0.1	OpenFl…	178	Type: OFPT_PACKET_IN
14	4.100791096	127.0.0.1	127.0.0.1	OpenFl…	176	Type: OFPT_PACKET_OUT
16	4.102673693	127.0.0.1	127.0.0.1	OpenFl…	178	Type: OFPT_PACKET_IN
17	4.103548052	127.0.0.1	127.0.0.1	OpenFl…	176	Type: OFPT_PACKET_OUT
19	5.118613804	127.0.0.1	127.0.0.1	OpenFl…	178	Type: OFPT_PACKET_IN
20	5.119664968	127.0.0.1	127.0.0.1	OpenFl…	176	Type: OFPT_PACKET_OUT
22	5.630883037	127.0.0.1	127.0.0.1	OpenFl…	178	Type: OFPT_PACKET_IN
23	5.632230020	127.0.0.1	127.0.0.1	OpenFl…	176	Type: OFPT_PACKET_OUT
25	5.634490681	127.0.0.1	127.0.0.1	OpenFl…	178	Type: OFPT_PACKET_IN
26	5.635740940	127.0.0.1	127.0.0.1	OpenFl…	176	Type: OFPT_PACKET_OUT
55	20.479150223	127.0.0.1	127.0.0.1	OpenFl…	178	Type: OFPT_PACKET_IN
56	20.480495853	127.0.0.1	127.0.0.1	OpenFl…	176	Type: OFPT_PACKET_OUT
151	57.667056775	127.0.0.1	127.0.0.1	OpenFl…	206	Type: OFPT_PACKET_IN
153	57.667591785	127.0.0.1	127.0.0.1	OpenFl…	206	Type: OFPT_PACKET_IN
155	57.668120763	127.0.0.1	127.0.0.1	OpenFl…	196	Type: OFPT_PACKET_OUT
158	57.669023964	127.0.0.1	127.0.0.1	OpenFl…	196	Type: OFPT_PACKET_OUT
160	57.673791224	127.0.0.1	127.0.0.1	OpenFl…	198	Type: OFPT_PACKET_IN
161	57.674589756	127.0.0.1	127.0.0.1	OpenFl…	196	Type: OFPT_PACKET_OUT
163	58.134437193	127.0.0.1	127.0.0.1	OpenFl…	206	Type: OFPT_PACKET_IN
164	58.136402792	127.0.0.1	127.0.0.1	OpenFl…	196	Type: OFPT_PACKET_OUT
166	58.239770898	127.0.0.1	127.0.0.1	OpenFl…	194	Type: OFPT_PACKET_IN
167	58.241799665	127.0.0.1	127.0.0.1	OpenFl…	192	Type: OFPT_PACKET_OUT

```
Frame 1: Packet, 178 bytes on wire (1424 bits), 178 bytes captured (1424 bits) on interface lo, id 0
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 48816, Dst Port: 6633, Seq: 1, Ack: 1, Len: 112
OpenFlow 1.3
```

When it is up:



openflow_v4

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	127.0.0.1	127.0.0.1	OpenFl…	178	Type: OFPT_PACKET_IN
2	0.001344989	127.0.0.1	127.0.0.1	OpenFl…	176	Type: OFPT_PACKET_OUT
4	0.003498315	127.0.0.1	127.0.0.1	OpenFl…	178	Type: OFPT_PACKET_IN
5	0.003665794	127.0.0.1	127.0.0.1	OpenFl…	178	Type: OFPT_PACKET_IN
6	0.004867149	127.0.0.1	127.0.0.1	OpenFl…	176	Type: OFPT_PACKET_OUT
8	0.007002835	127.0.0.1	127.0.0.1	OpenFl…	176	Type: OFPT_PACKET_OUT
10	4.095795420	127.0.0.1	127.0.0.1	OpenFl…	178	Type: OFPT_PACKET_IN
11	4.096852031	127.0.0.1	127.0.0.1	OpenFl…	176	Type: OFPT_PACKET_OUT
13	4.099977599	127.0.0.1	127.0.0.1	OpenFl…	178	Type: OFPT_PACKET_IN
14	4.100791096	127.0.0.1	127.0.0.1	OpenFl…	176	Type: OFPT_PACKET_OUT
16	4.102673693	127.0.0.1	127.0.0.1	OpenFl…	178	Type: OFPT_PACKET_IN
17	4.103548052	127.0.0.1	127.0.0.1	OpenFl…	176	Type: OFPT_PACKET_OUT
19	5.118613804	127.0.0.1	127.0.0.1	OpenFl…	178	Type: OFPT_PACKET_IN
20	5.119664968	127.0.0.1	127.0.0.1	OpenFl…	176	Type: OFPT_PACKET_OUT
22	5.630883037	127.0.0.1	127.0.0.1	OpenFl…	178	Type: OFPT_PACKET_IN
23	5.632230020	127.0.0.1	127.0.0.1	OpenFl…	176	Type: OFPT_PACKET_OUT
25	5.634490681	127.0.0.1	127.0.0.1	OpenFl…	178	Type: OFPT_PACKET_IN
26	5.635740940	127.0.0.1	127.0.0.1	OpenFl…	176	Type: OFPT_PACKET_OUT
28	6.096858178	127.0.0.1	127.0.0.1	OpenFl…	300	Type: OFPT_PACKET_IN
29	6.097487412	127.0.0.1	127.0.0.1	OpenFl…	298	Type: OFPT_PACKET_OUT
31	6.144179437	127.0.0.1	127.0.0.1	OpenFl…	300	Type: OFPT_PACKET_IN
32	6.144782415	127.0.0.1	127.0.0.1	OpenFl…	298	Type: OFPT_PACKET_OUT
34	6.146421037	127.0.0.1	127.0.0.1	OpenFl…	300	Type: OFPT_PACKET_IN
35	6.147010677	127.0.0.1	127.0.0.1	OpenFl…	298	Type: OFPT_PACKET_OUT
37	11.098524145	127.0.0.1	127.0.0.1	OpenFl…	74	Type: OFPT_ECHO_REQUEST
38	11.099110046	127.0.0.1	127.0.0.1	OpenFl…	74	Type: OFPT_ECHO_REPLY
40	11.145771766	127.0.0.1	127.0.0.1	OpenFl…	74	Type: OFPT_ECHO_REQUEST
41	11.146382319	127.0.0.1	127.0.0.1	OpenFl…	74	Type: OFPT_ECHO_REPLY
43	11.148364504	127.0.0.1	127.0.0.1	OpenFl…	74	Type: OFPT_ECHO_REQUEST
44	11.148996116	127.0.0.1	127.0.0.1	OpenFl…	74	Type: OFPT_ECHO_REPLY

```
Frame 1: Packet, 178 bytes on wire (1424 bits), 178 bytes captured (1424 bits) on interface lo, id 0
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 48816, Dst Port: 6633, Seq: 1, Ack: 1, Len: 112
OpenFlow 1.3
```

openflow_v4

No.	Time	Source	Destination	Protocol	Length	Info
558	118.389420170	127.0.0.1	127.0.0.1	OpenFl…	180	Type: OFPT_PACKET_OUT
560	118.390966604	127.0.0.1	127.0.0.1	OpenFl…	182	Type: OFPT_PACKET_IN
561	118.391875184	127.0.0.1	127.0.0.1	OpenFl…	170	Type: OFPT_FLOW_MOD
563	118.391951993	127.0.0.1	127.0.0.1	OpenFl…	180	Type: OFPT_PACKET_OUT
565	118.393439125	127.0.0.1	127.0.0.1	OpenFl…	182	Type: OFPT_PACKET_IN
566	118.394317655	127.0.0.1	127.0.0.1	OpenFl…	170	Type: OFPT_FLOW_MOD
568	118.394374086	127.0.0.1	127.0.0.1	OpenFl…	180	Type: OFPT_PACKET_OUT
570	118.398983476	127.0.0.1	127.0.0.1	OpenFl…	182	Type: OFPT_PACKET_IN
571	118.400394251	127.0.0.1	127.0.0.1	OpenFl…	170	Type: OFPT_FLOW_MOD
572	118.400451252	127.0.0.1	127.0.0.1	OpenFl…	180	Type: OFPT_PACKET_OUT
574	118.402078868	127.0.0.1	127.0.0.1	OpenFl…	182	Type: OFPT_PACKET_IN
575	118.402972305	127.0.0.1	127.0.0.1	OpenFl…	170	Type: OFPT_FLOW_MOD
576	118.402917935	127.0.0.1	127.0.0.1	OpenFl…	180	Type: OFPT_PACKET_OUT
578	118.404563275	127.0.0.1	127.0.0.1	OpenFl…	182	Type: OFPT_PACKET_IN
579	118.405630307	127.0.0.1	127.0.0.1	OpenFl…	170	Type: OFPT_FLOW_MOD
580	118.405686802	127.0.0.1	127.0.0.1	OpenFl…	180	Type: OFPT_PACKET_OUT
582	122.386813058	127.0.0.1	127.0.0.1	OpenFl…	300	Type: OFPT_PACKET_IN
583	122.387537138	127.0.0.1	127.0.0.1	OpenFl…	298	Type: OFPT_PACKET_OUT
584	122.389076825	127.0.0.1	127.0.0.1	OpenFl…	300	Type: OFPT_PACKET_IN
585	122.389756263	127.0.0.1	127.0.0.1	OpenFl…	298	Type: OFPT_PACKET_OUT
588	122.496435711	127.0.0.1	127.0.0.1	OpenFl…	300	Type: OFPT_PACKET_IN
589	122.497131752	127.0.0.1	127.0.0.1	OpenFl…	298	Type: OFPT_PACKET_OUT
591	122.878426893	127.0.0.1	127.0.0.1	OpenFl…	178	Type: OFPT_PACKET_IN
592	122.879130116	127.0.0.1	127.0.0.1	OpenFl…	176	Type: OFPT_PACKET_OUT
594	126.974364225	127.0.0.1	127.0.0.1	OpenFl…	178	Type: OFPT_PACKET_IN
595	126.975114516	127.0.0.1	127.0.0.1	OpenFl…	176	Type: OFPT_PACKET_OUT
597	126.976657754	127.0.0.1	127.0.0.1	OpenFl…	178	Type: OFPT_PACKET_IN
598	126.977295448	127.0.0.1	127.0.0.1	OpenFl…	176	Type: OFPT_PACKET_OUT
600	127.880117528	127.0.0.1	127.0.0.1	OpenFl…	74	Type: OFPT_ECHO_REQUEST
601	127.880346155	127.0.0.1	127.0.0.1	OpenFl…	74	Type: OFPT_ECHO_REPLY

```
▸ Frame 1: Packet, 178 bytes on wire (1424 bits), 178 bytes captured (1424 bits) on interface lo, id 0
▸ Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
▸ Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
▸ Transmission Control Protocol, Src Port: 48816, Dst Port: 6633, Seq: 1, Ack: 1, Len: 112
▸ OpenFlow 1.3
```

– Scenario 2: Port Failure (OVS Statistics)

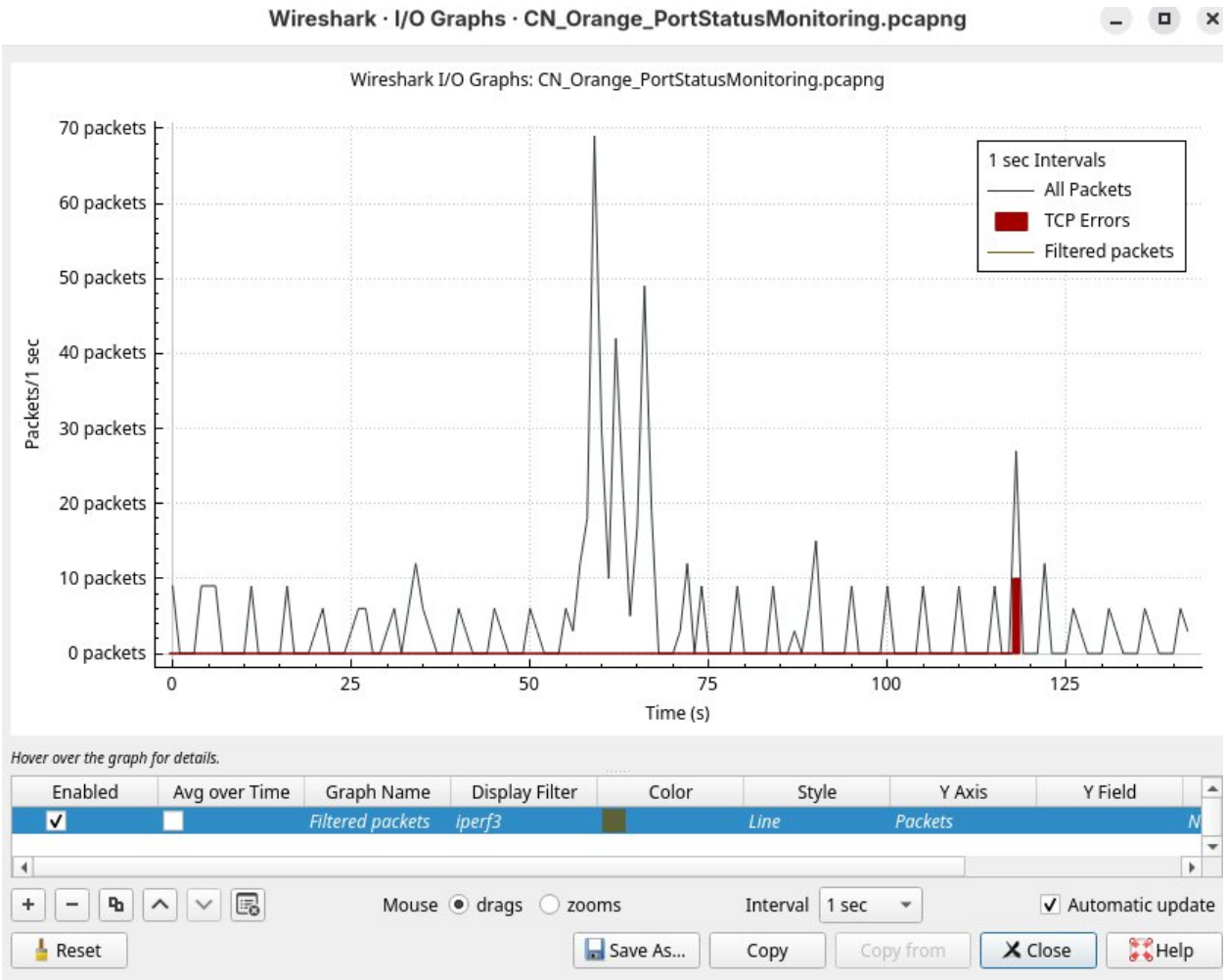
dump-ports s101

No.	Time	Source	Destination	Protocol	Length	Info
183	59.269947633	127.0.0.1	127.0.0.1	OpenFl…	196	Type: OFPT_PACKET_OUT
185	59.271557446	127.0.0.1	127.0.0.1	OpenFl…	246	Type: OFPT_PACKET_IN
186	59.272421409	127.0.0.1	127.0.0.1	OpenFl…	198	Type: OFPT_PACKET_IN
187	59.272943547	127.0.0.1	127.0.0.1	OpenFl…	176	Type: OFPT_PACKET_OUT
188	59.274546074	127.0.0.1	127.0.0.1	OpenFl…	178	Type: OFPT_PACKET_IN
190	59.275803484	127.0.0.1	127.0.0.1	OpenFl…	236	Type: OFPT_PACKET_OUT
192	59.276444463	127.0.0.1	127.0.0.1	OpenFl…	196	Type: OFPT_PACKET_OUT
194	59.276875019	127.0.0.1	127.0.0.1	OpenFl…	176	Type: OFPT_PACKET_OUT
196	59.277460280	127.0.0.1	127.0.0.1	OpenFl…	238	Type: OFPT_PACKET_IN
197	59.278483303	127.0.0.1	127.0.0.1	OpenFl…	236	Type: OFPT_PACKET_OUT
198	59.306328091	127.0.0.1	127.0.0.1	OpenFl…	266	Type: OFPT_PACKET_IN
199	59.307677914	127.0.0.1	127.0.0.1	OpenFl…	256	Type: OFPT_PACKET_OUT
200	59.309323543	127.0.0.1	127.0.0.1	OpenFl…	258	Type: OFPT_PACKET_IN
201	59.310992339	127.0.0.1	127.0.0.1	OpenFl…	256	Type: OFPT_PACKET_OUT
204	59.368387942	127.0.0.1	127.0.0.1	OpenFl…	300	Type: OFPT_PACKET_IN
205	59.369128791	127.0.0.1	127.0.0.1	OpenFl…	298	Type: OFPT_PACKET_OUT
207	59.370752312	127.0.0.1	127.0.0.1	OpenFl…	300	Type: OFPT_PACKET_IN
208	59.371488219	127.0.0.1	127.0.0.1	OpenFl…	298	Type: OFPT_PACKET_OUT
210	59.391845180	127.0.0.1	127.0.0.1	OpenFl…	206	Type: OFPT_PACKET_IN
211	59.391918402	127.0.0.1	127.0.0.1	OpenFl…	178	Type: OFPT_PACKET_IN
213	59.394280288	127.0.0.1	127.0.0.1	OpenFl…	226	Type: OFPT_PACKET_IN
214	59.397036115	127.0.0.1	127.0.0.1	OpenFl…	196	Type: OFPT_PACKET_OUT
216	59.399456828	127.0.0.1	127.0.0.1	OpenFl…	176	Type: OFPT_PACKET_OUT
218	59.400011921	127.0.0.1	127.0.0.1	OpenFl…	216	Type: OFPT_PACKET_OUT
220	59.452101342	127.0.0.1	127.0.0.1	OpenFl…	320	Type: OFPT_PACKET_IN
221	59.452827108	127.0.0.1	127.0.0.1	OpenFl…	318	Type: OFPT_PACKET_OUT
223	59.454445866	127.0.0.1	127.0.0.1	OpenFl…	320	Type: OFPT_PACKET_IN
224	59.455223419	127.0.0.1	127.0.0.1	OpenFl…	318	Type: OFPT_PACKET_OUT
226	59.492515036	127.0.0.1	127.0.0.1	OpenFl…	300	Type: OFPT_PACKET_IN
227	59.493368938	127.0.0.1	127.0.0.1	OpenFl…	298	Type: OFPT_PACKET_OUT

```
▸ Frame 1: Packet, 178 bytes on wire (1424 bits), 178 bytes captured (1424 bits) on interface lo, id 0
▸ Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
▸ Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
▸ Transmission Control Protocol, Src Port: 48816, Dst Port: 6633, Seq: 1, Ack: 1, Len: 112
▸ OpenFlow 1.3
```

dump-flows s101

No.	Time	Source	Destination	Protocol	Length	Info
88	33.976622969	127.0.0.1	127.0.0.1	OpenFl…	972	Type: OFPT_PACKET_IN
89	33.977561269	127.0.0.1	127.0.0.1	OpenFl…	972	Type: OFPT_PACKET_IN
90	33.979387636	127.0.0.1	127.0.0.1	OpenFl…	970	Type: OFPT_PACKET_OUT
92	33.982892868	127.0.0.1	127.0.0.1	OpenFl…	970	Type: OFPT_PACKET_OUT
94	34.339423229	127.0.0.1	127.0.0.1	OpenFl…	972	Type: OFPT_PACKET_IN
95	34.339571919	127.0.0.1	127.0.0.1	OpenFl…	972	Type: OFPT_PACKET_IN
96	34.342202138	127.0.0.1	127.0.0.1	OpenFl…	970	Type: OFPT_PACKET_OUT
98	34.342283177	127.0.0.1	127.0.0.1	OpenFl…	970	Type: OFPT_PACKET_OUT
100	34.800360254	127.0.0.1	127.0.0.1	OpenFl…	972	Type: OFPT_PACKET_IN
101	34.801937878	127.0.0.1	127.0.0.1	OpenFl…	970	Type: OFPT_PACKET_OUT
103	34.914678288	127.0.0.1	127.0.0.1	OpenFl…	972	Type: OFPT_PACKET_IN
104	34.917166733	127.0.0.1	127.0.0.1	OpenFl…	970	Type: OFPT_PACKET_OUT
106	35.422923064	127.0.0.1	127.0.0.1	OpenFl…	972	Type: OFPT_PACKET_IN
107	35.424886815	127.0.0.1	127.0.0.1	OpenFl…	970	Type: OFPT_PACKET_OUT
109	35.540496996	127.0.0.1	127.0.0.1	OpenFl…	972	Type: OFPT_PACKET_IN
110	35.542040812	127.0.0.1	127.0.0.1	OpenFl…	970	Type: OFPT_PACKET_OUT
112	36.106052005	127.0.0.1	127.0.0.1	OpenFl…	74	Type: OFPT_ECHO_REQUEST
113	36.106598775	127.0.0.1	127.0.0.1	OpenFl…	74	Type: OFPT_ECHO_REPLY
115	40.426394318	127.0.0.1	127.0.0.1	OpenFl…	74	Type: OFPT_ECHO_REQUEST
116	40.427046417	127.0.0.1	127.0.0.1	OpenFl…	74	Type: OFPT_ECHO_REPLY
118	40.543035378	127.0.0.1	127.0.0.1	OpenFl…	74	Type: OFPT_ECHO_REQUEST
119	40.543588648	127.0.0.1	127.0.0.1	OpenFl…	74	Type: OFPT_ECHO_REPLY
121	41.106673719	127.0.0.1	127.0.0.1	OpenFl…	74	Type: OFPT_ECHO_REQUEST
122	41.107227662	127.0.0.1	127.0.0.1	OpenFl…	74	Type: OFPT_ECHO_REPLY
124	45.428195155	127.0.0.1	127.0.0.1	OpenFl…	74	Type: OFPT_ECHO_REQUEST
125	45.428785025	127.0.0.1	127.0.0.1	OpenFl…	74	Type: OFPT_ECHO_REPLY
127	45.545356068	127.0.0.1	127.0.0.1	OpenFl…	74	Type: OFPT_ECHO_REQUEST
128	45.545846898	127.0.0.1	127.0.0.1	OpenFl…	74	Type: OFPT_ECHO_REPLY
130	46.107997245	127.0.0.1	127.0.0.1	OpenFl…	74	Type: OFPT_ECHO_REQUEST
131	46.108517037	127.0.0.1	127.0.0.1	OpenFl…	74	Type: OFPT_ECHO_REPLY

```
▸ Frame 1: Packet, 178 bytes on wire (1424 bits), 178 bytes captured (1424 bits) on interface lo, id 0
▸ Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
▸ Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
▸ Transmission Control Protocol, Src Port: 48816, Dst Port: 6633, Seq: 1, Ack: 1, Len: 112
▸ OpenFlow 1.3
```

Scenario 3: Throughput